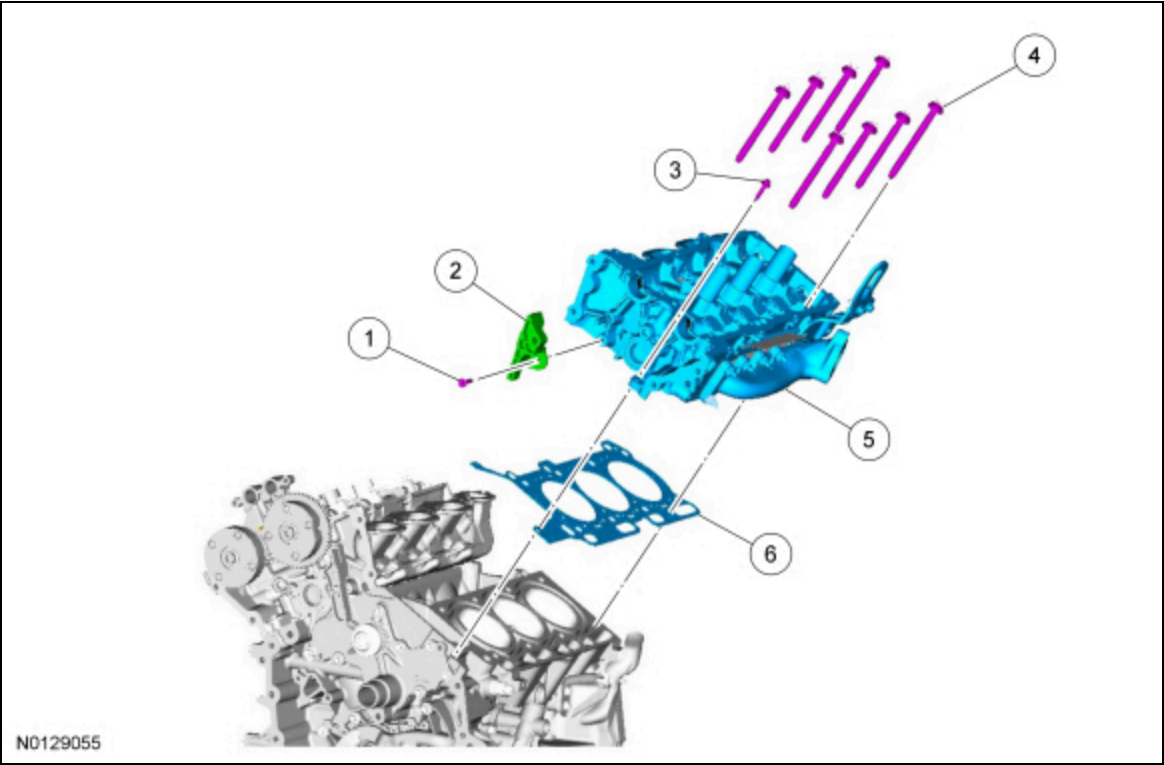


Cylinder Head — LH

 [Ford Ecat Electronic Parts Catalog](#)

Material

Item	Specification
Motorcraft® Metal Surface Prep Wipes ZC-31-B	—
Motorcraft® SAE 5W-30 Synthetic Blend Motor Oil (US); Motorcraft® SAE 5W-30 Super Premium Motor Oil (Canada) XO-5W30-QSP (US); CXO-5W30-LSP12 (Canada)	WSS- M2C946-A
Motorcraft® Silicone Gasket Remover ZC-30-A	—



Item	Part Number	Description
1	W709818 W709818	LH upper primary timing chain guide bolt
2	6K297 6K297	LH upper primary timing chain guide
3	W710702 W710702	Cylinder head M6 bolt
4	6065 6065	Cylinder head bolt (8 required)
5	6049 6049	Cylinder head
6	6051 6051	Cylinder head gasket

Removal

WARNING: Do not smoke, carry lighted tobacco or have an open flame of any type when working on or near any fuel-related component. Highly flammable mixtures are always present and may be ignited. Failure to follow these instructions may result in serious personal injury.

WARNING: Before working on or disconnecting any of the fuel tubes or fuel system components, relieve the fuel system pressure to prevent accidental spraying of fuel. Fuel in the fuel system remains under high pressure, even when the engine is not running. Failure to follow this instruction may result in serious personal injury.

WARNING: Do not carry personal electronic devices such as cell phones, pagers or audio equipment of any type when working on or near any fuel-related component. Highly flammable mixtures are always present and may be ignited. Failure to follow these instructions may result in serious personal injury.

WARNING: Always disconnect the battery ground cable at the battery when working on an evaporative emission (EVAP) system or fuel-related component. Highly flammable mixtures are always present and may be ignited. Failure to follow these instructions may result in serious personal injury.

WARNING: Clean all fuel residue from the engine compartment. If not removed, fuel residue may ignite when the engine is returned to operation. Failure to follow this instruction may result in serious personal injury.

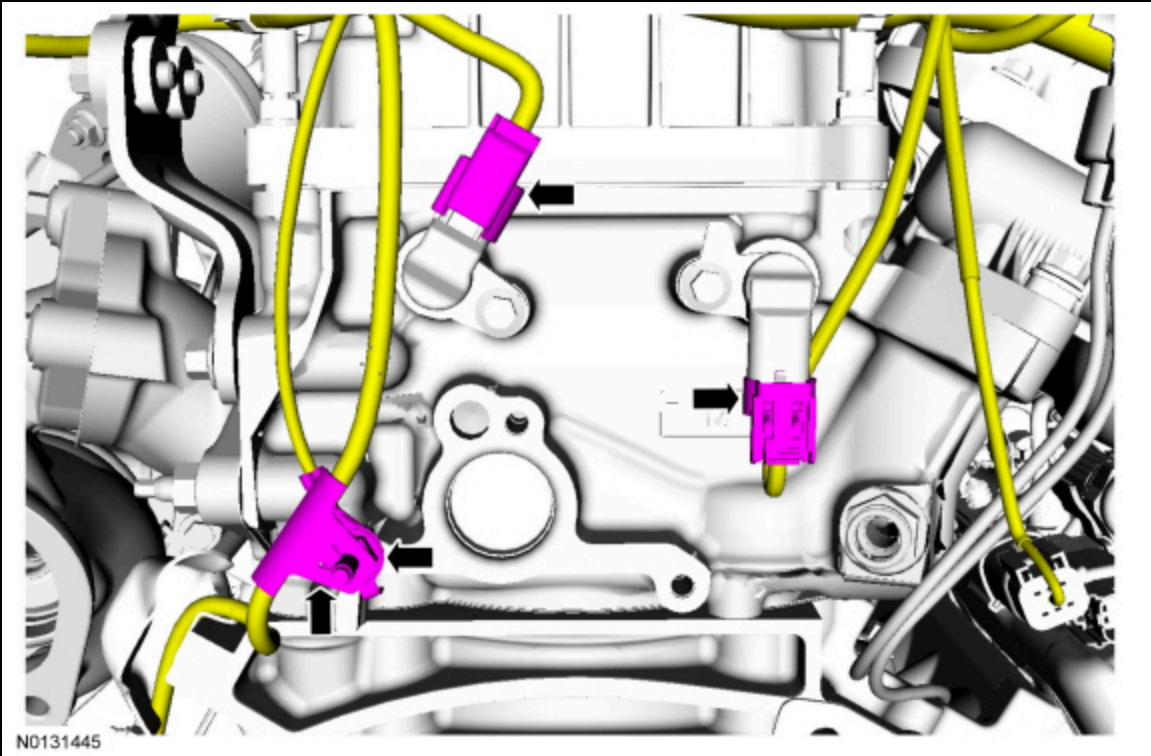
NOTICE: During engine repair procedures, cleanliness is extremely important. Any foreign material, including any material created while cleaning gasket surfaces that enters the oil passages, coolant passages or the oil pan, may cause engine failure.

NOTE: *If the cylinder head is replaced, a new secondary timing chain tensioner will need to be installed.*

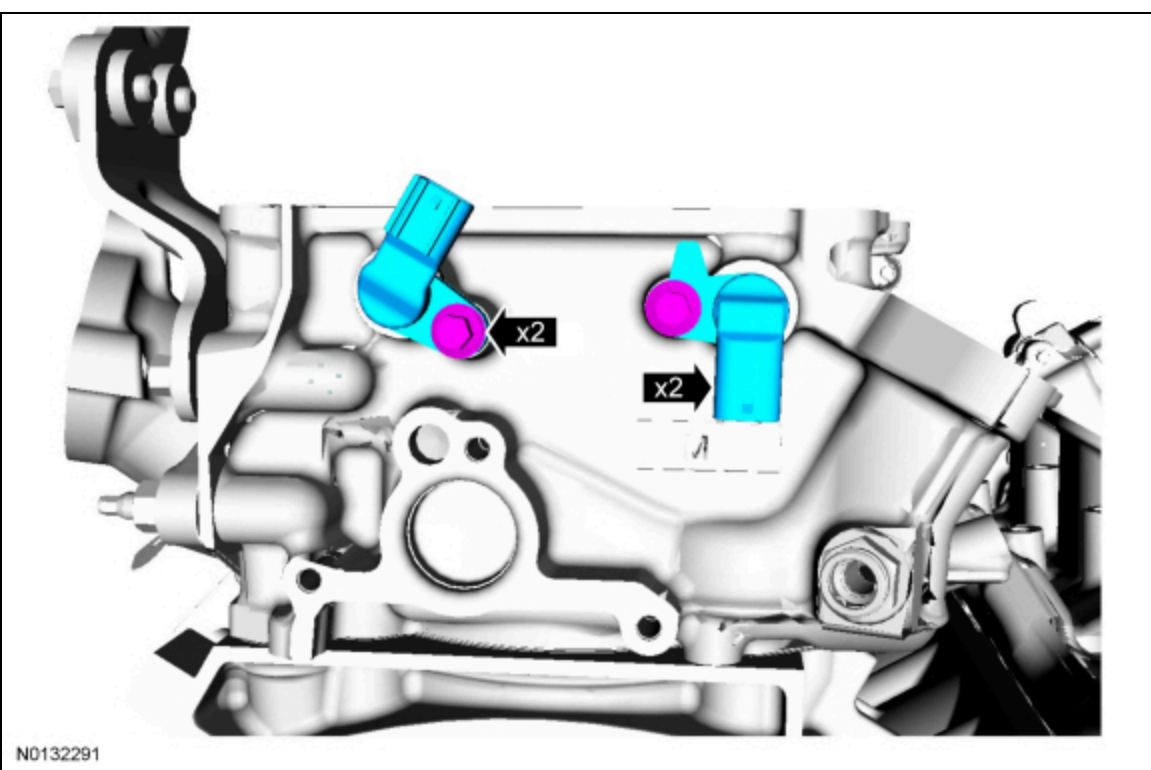
1. Release the fuel pressure. For additional information, refer to Section 310-00.
2. Remove the LH turbocharger. For additional information, refer to Section 303-04E.
3. Remove the LH camshafts. For additional information, refer to Camshaft — LH in this section.
4. Remove the fuel rail. For additional information, refer to Section 303-04A.
5. **NOTICE:** Do not use power tools to remove the bolt or damage to the LH primary timing chain guide may occur.

Remove the bolt and the LH upper primary timing chain guide.

6. Disconnect the 2 LH Camshaft Position (CMP) sensor electrical connectors. Position back the Crankshaft Position (CKP) wire retainer. Remove the ground stud bolt.

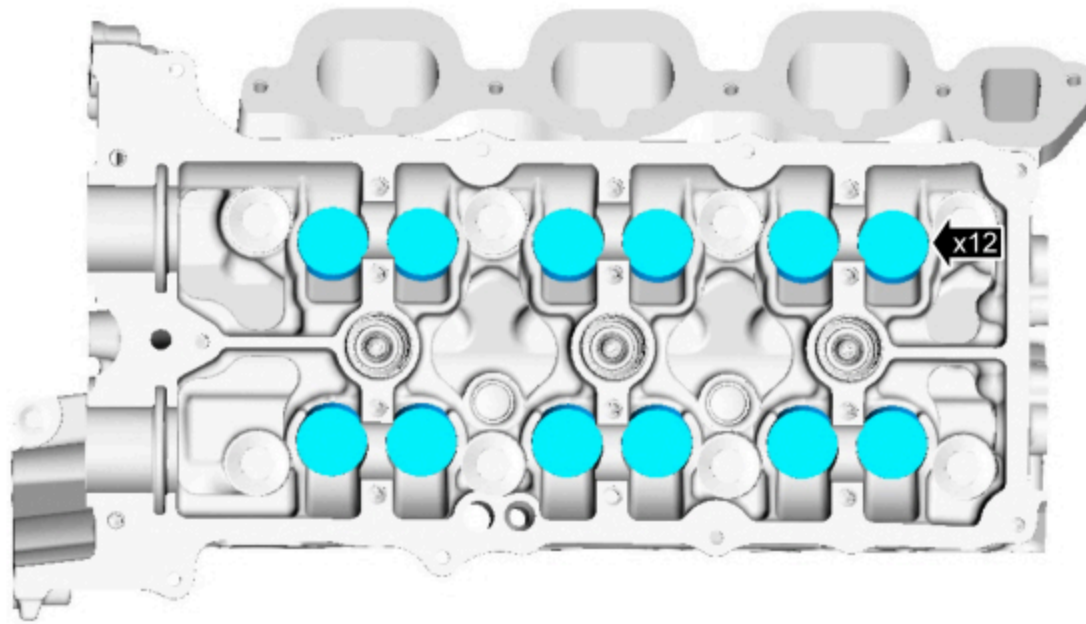


7. Remove the 2 bolts and the 2 LH CMP sensors.



8. **NOTE:** If the components are to be reinstalled, they must be installed in the same positions. Mark the components for installation into their original positions.

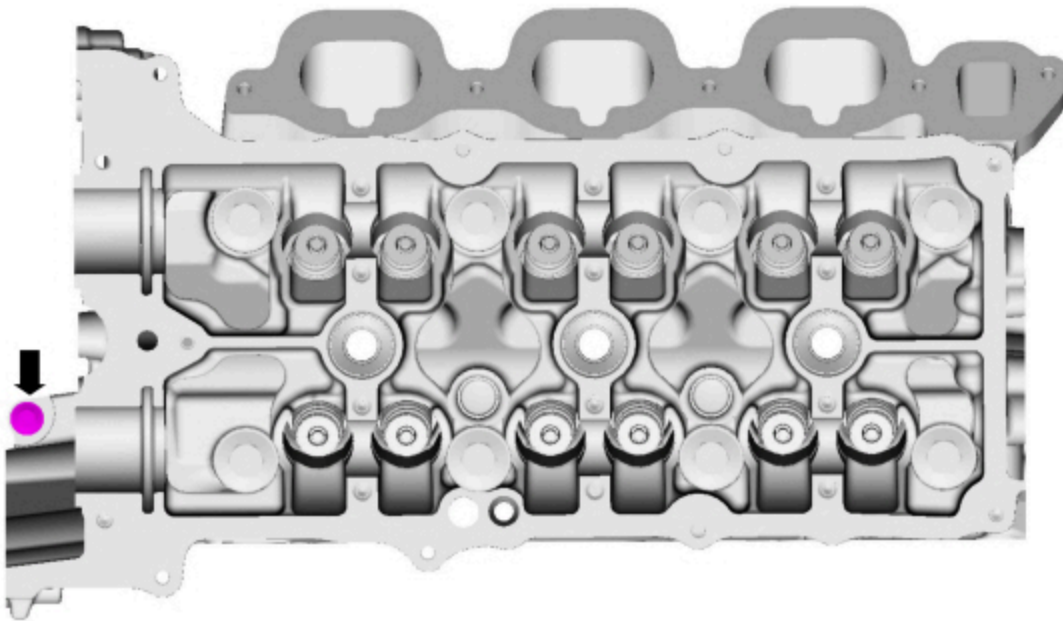
Remove the valve tappets from the cylinder head.



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9. Inspect the valve tappets. For additional information, refer to Section 303-00.

10. Remove and discard the M6 bolt.



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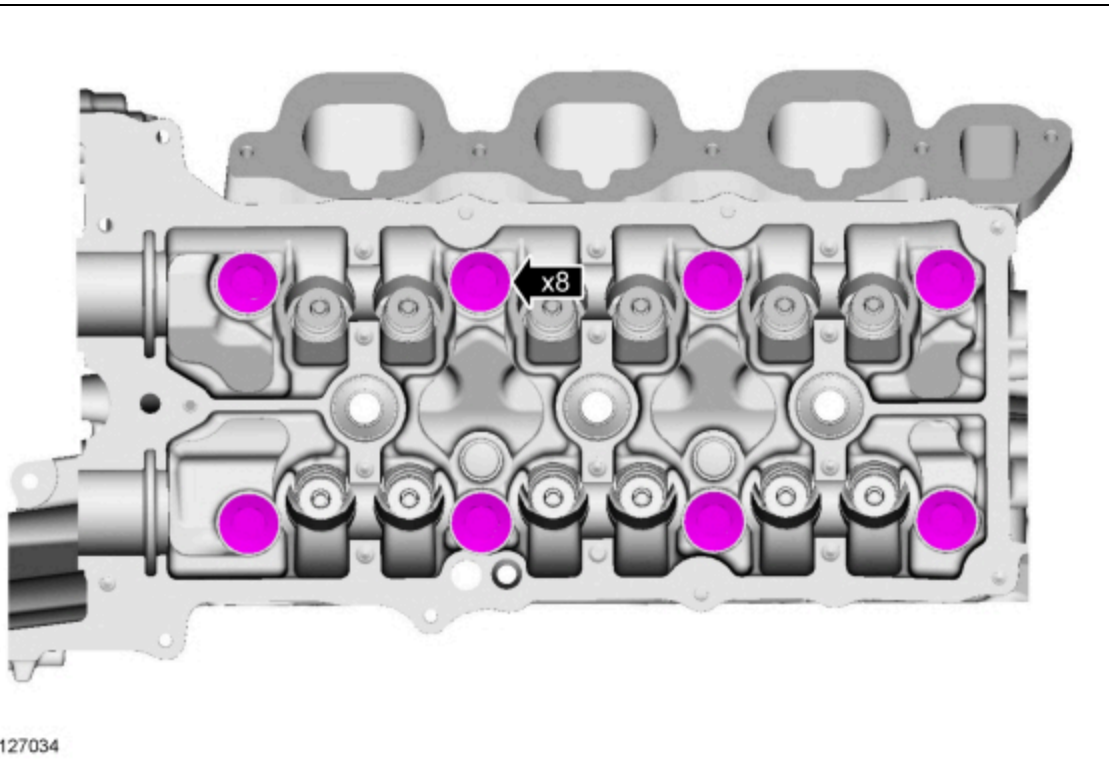
11. **NOTICE:** Place clean, lint-free shop towels over exposed engine cavities. Carefully remove the towels so foreign material is not dropped into the engine. Any foreign material (including any material created while cleaning gasket surfaces) that enters the oil passages or the oil pan, may cause engine failure

NOTICE: Aluminum surfaces are soft and may be scratched easily. Never place the cylinder head gasket surface, unprotected, on a bench surface.

NOTE: The cylinder head bolts must be discarded and new bolts must be installed. They are a tighten-to-yield design and cannot be reused.

Remove and discard the 8 bolts from the cylinder head.

- Remove the cylinder head.
- Discard the cylinder head gasket.



12. **NOTICE:** Do not use metal scrapers, wire brushes, power abrasive discs or other abrasive means to clean the sealing surfaces. These tools cause scratches and gouges that make leak paths. Use a plastic scraping tool to remove all traces of the head gasket.

NOTE: Observe all warnings or cautions and follow all application directions contained on the packaging of the silicone gasket remover and the metal surface prep.

NOTE: If there is no residual gasket material present, metal surface prep can be used to clean and prepare the surfaces.

Clean the cylinder head-to-cylinder block mating surfaces of both the cylinder heads and the cylinder block in the following sequence.

1. Remove any large deposits of silicone or gasket material with a plastic scraper.
2. Apply silicone gasket remover, following package directions, and allow to set for several minutes.
3. Remove the silicone gasket remover with a plastic scraper. A second application of silicone gasket remover may be required if residual traces of silicone or gasket material remain.
4. Apply metal surface prep, following package directions, to remove any remaining traces of oil or coolant and to prepare the surfaces to bond with the new gasket. Do not attempt to make the metal shiny. Some staining of the metal surfaces is normal

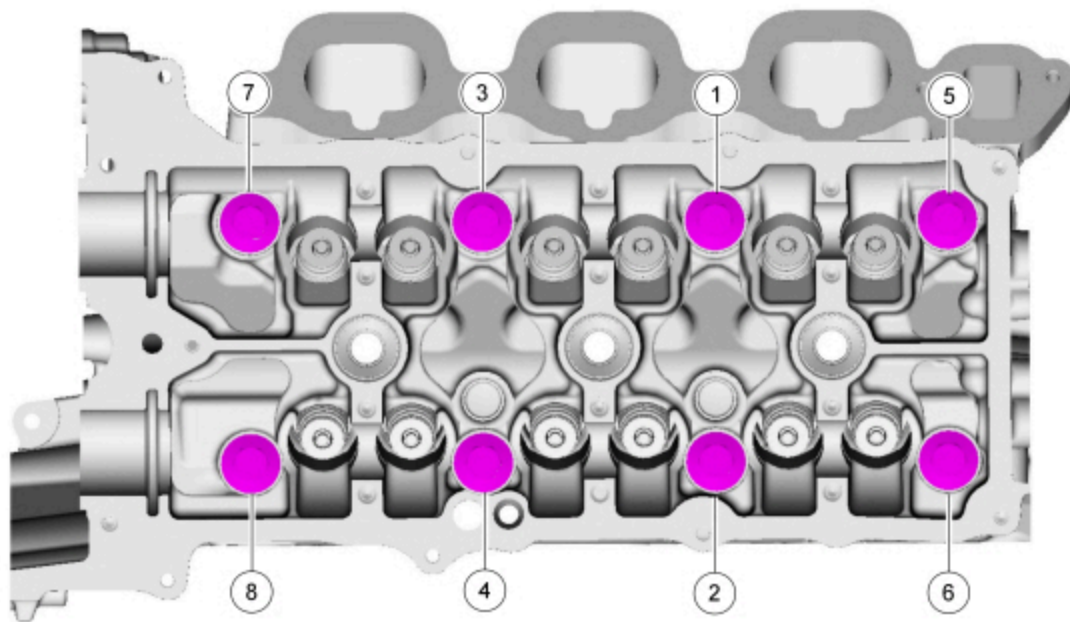
13. Support the cylinder head on a bench with the head gasket side up. Check the cylinder head distortion and the cylinder block distortion. For additional information, refer to Section 303-00.

Installation

1. **NOTE:** If the cylinder head is replaced, a new secondary timing chain tensioner will need to be installed.

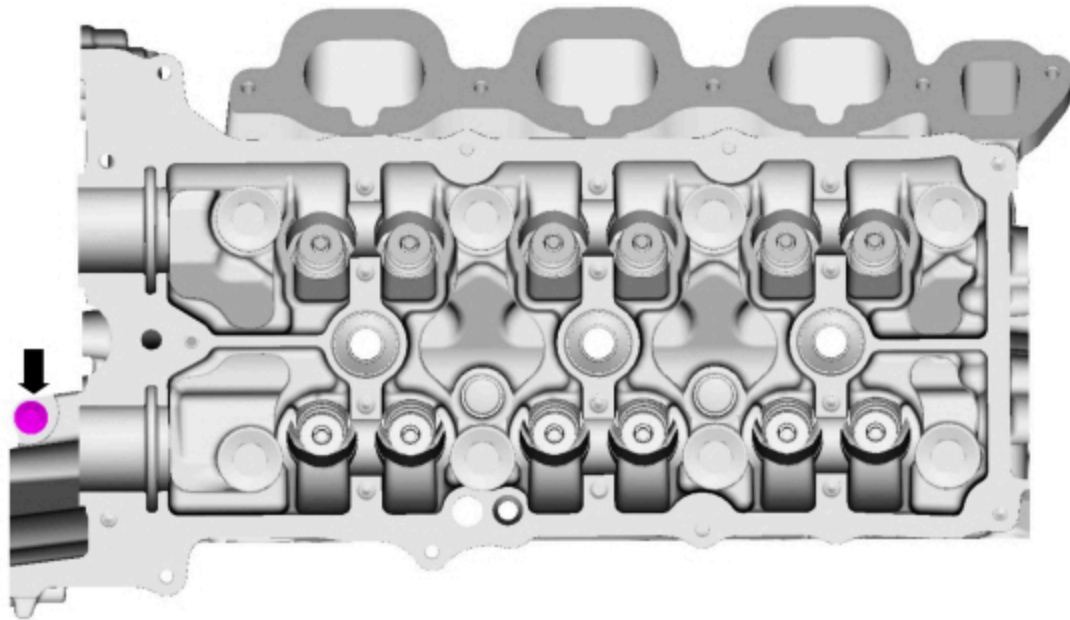
Install a new gasket, the LH cylinder head and 8 new bolts. Tighten in the sequence shown in 5 stages:

- Stage 1: Tighten to 20 Nm (177 lb-in).
- Stage 2: Tighten to 35 Nm (26 lb-ft).
- Stage 3: Tighten 90 degrees.
- Stage 4: Tighten 90 degrees
- Stage 5: Tighten 45 degrees.



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2. Install the new M6 bolt.
 - Tighten to 10 Nm (89 lb-in).

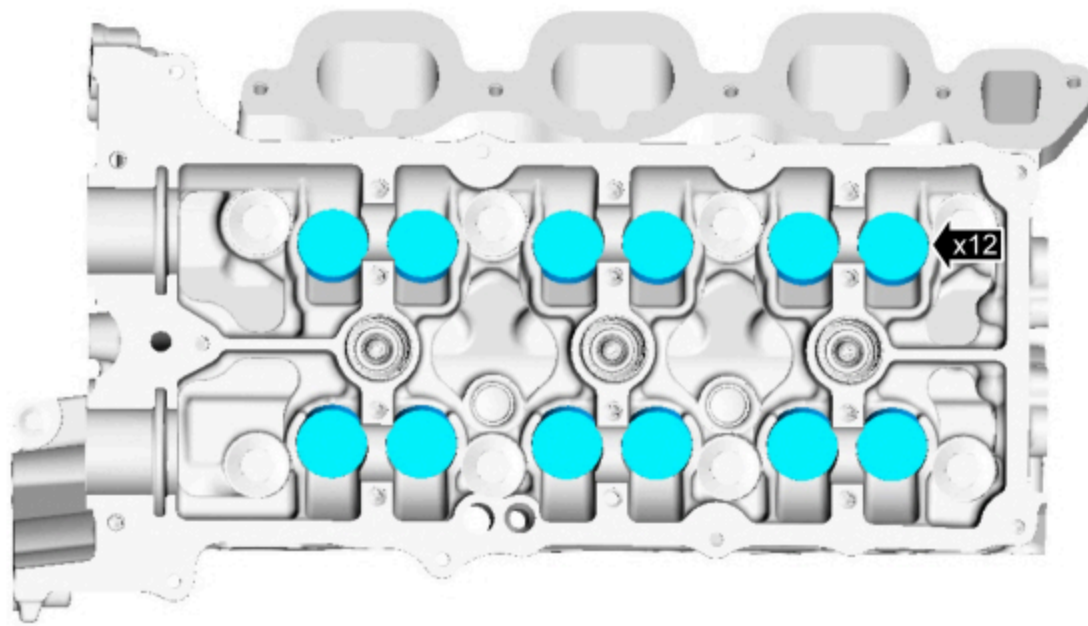


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3. **NOTE:** *The valve tappets must be installed in their original positions.*

NOTE: *Coat the valve tappets with clean engine oil prior to installation.*

Install the valve tappets.

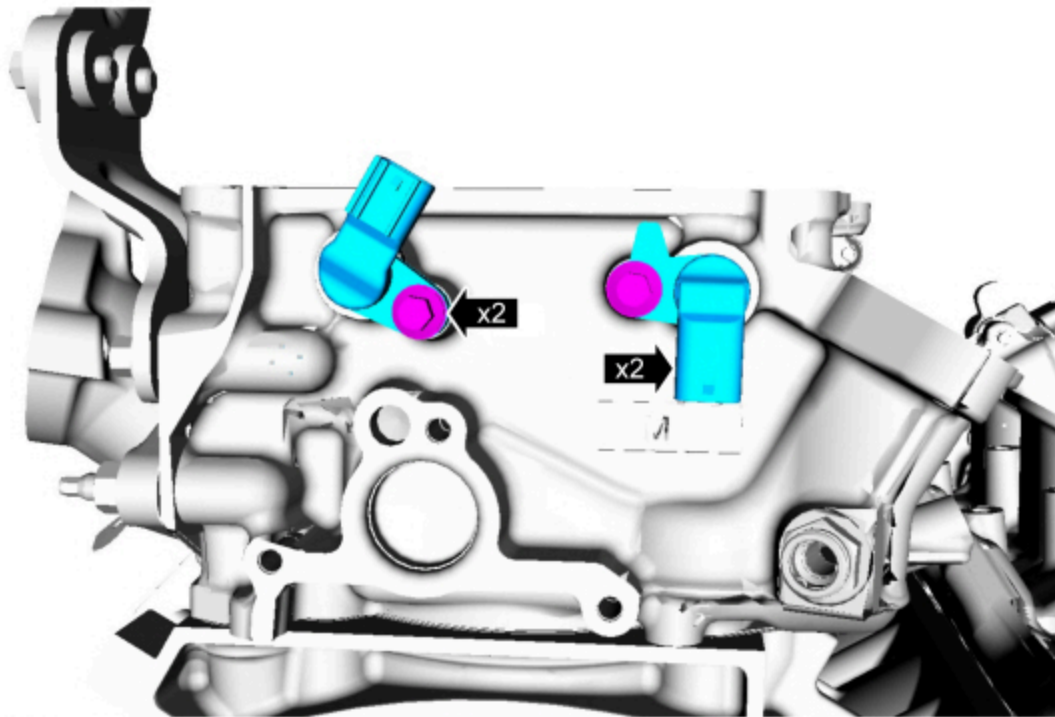


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4. **NOTE:** Lubricate the 2 CMP sensor O-ring seals with clean engine oil.

Install the 2 LH CMP sensors and the 2 bolts.

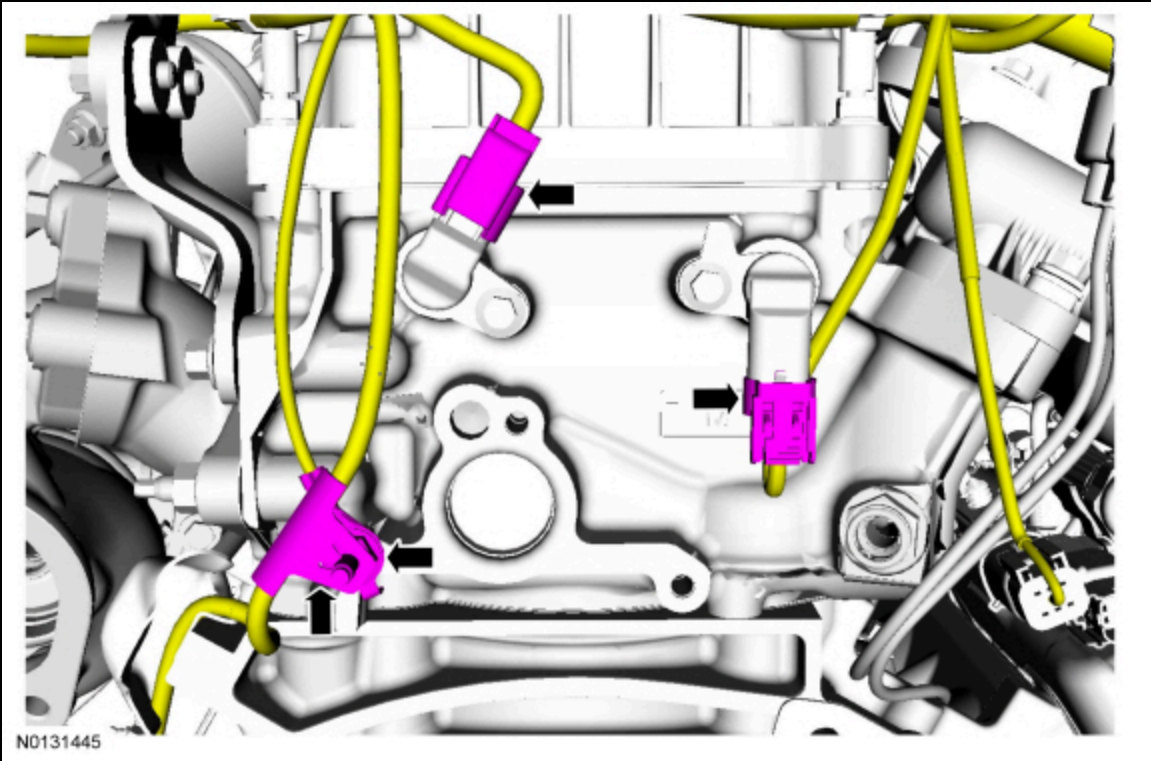
- Tighten to 10 Nm (89 lb-in).



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5. Install the ground stud bolt. Position back the CKP wire retainer. Connect the 2 LH CMP sensor electrical connectors.

- Tighten to 10 Nm (89 lb-in) then an additional 45 degrees.



6. Install the LH upper primary timing chain guide and the bolt.
 - Tighten to 10 Nm (89 lb-in).
7. Install the fuel rail. For additional information, refer to Section 303-04A.
8. Install the LH camshafts. For additional information, refer to Camshaft — LH in this section.
9. Install the LH turbocharger. For additional information, refer to Section 303-04E.